

INT315:CLUSTER COMPUTING

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

- CO1 :: analyze the similarities and differences between spark and hadoop.
- CO2 :: discover the features and functions of scala language for apache spark.
- CO3 :: discuss the in-memory computation of RDDs for faster processing in spark
- CO4 :: apply spark mllib for performing machine learning in big data environment
- CO5 :: analyze workflows as well as processing and analysis of big data using apache spark
- CO6 :: apply spark graphx for performing graph processing in big data environment

Unit I

Introduction to spark : limitations of mapreduce in hadoop, comparison of batch vs. real-time analytics, application of stream processing and in-memory processing, features and benefits of spark, installation of spark as a standalone user, compare spark vs. hadoop ecosystem

Unit II

Introduction to programming in scala : features of scala, basic data types and literals used in scala, operators and methods used in scala, introduction to type inference, mutable vs. immutable collections, functions in scala, lists, maps and streams in scala

Unit III

Using RDD for creating applications in spark and Graph Analytics : Features of rdd, creating rdds, rdd functions, describe rdd operations and methods, invoking the spark shell, shared variables, Introduction to Spark GraphX, Spark GraphX Features, Spark GraphX Operations, Feature extraction and transformation

Unit IV

Running SQL queries using spark SQL : importance and features of sparksql, methods to convert rdds to dataframes, concepts of sparksql, joins, groupby, orderby, aggregate functions, SQL wildcards.

Unit V

Spark Streaming with Apache Kafka : Fundamentals of Apache Kafka, Apache Kafka Cluster Architecture, Apache Kafka Installation, Integration of Apache Kafka with Spark, producer & consumer messaging model, Kafka pipeline, ingestion patterns

Unit VI

Spark ML Programming and pyspark : Introduction to Machine Learning and pyspark, Techniques of Machine Learning, Introduction to Spark MLlib, Key Concepts of Spark ML, Spark ML Algorithms using pyspark, Linear Regression, Logistic Regression, Decision Tree, K-means, SVM, Naive Bayes, Evaluation and performance matrix-Confusion matrix, R2 , RMSE, MAE, Correlation heat map

List of Practicals / Experiments:

Practical List

- Demonstrate the use of mllib in apache spark
- Demonstrate the use of spark sql in apache spark
- Demonstrate the use of graphx in apache spark
- Demonstrate the use of spark streaming in apache spark
- Demonstrate the use of producer and consumer in apache kafka

Text Books:

1. ADVANCED ANALYTICS WITH PYSARK: PATTERNS FOR LEARNING FROM DATA AT SCALE USING PYTHON AND SPARK by AKASH TANDON, SHROFF/O'REILLY

References:

1. LEARNING SPARK by MATEI ZAHARIA, PATRICK WENDELL, ANDY KONWINSKI, HOLDEN KARAU, SHROFF/O'REILLY

